

Supporting Multiple Environments



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Module Overview



Multiple environments

- Development, QA, staging, production
- Same code

Multiple approaches

- Workspaces, tfvars files, code branches

Create multiple networking environments





Multiple Environment Approaches



Common Approaches

Terraform OSS workspaces

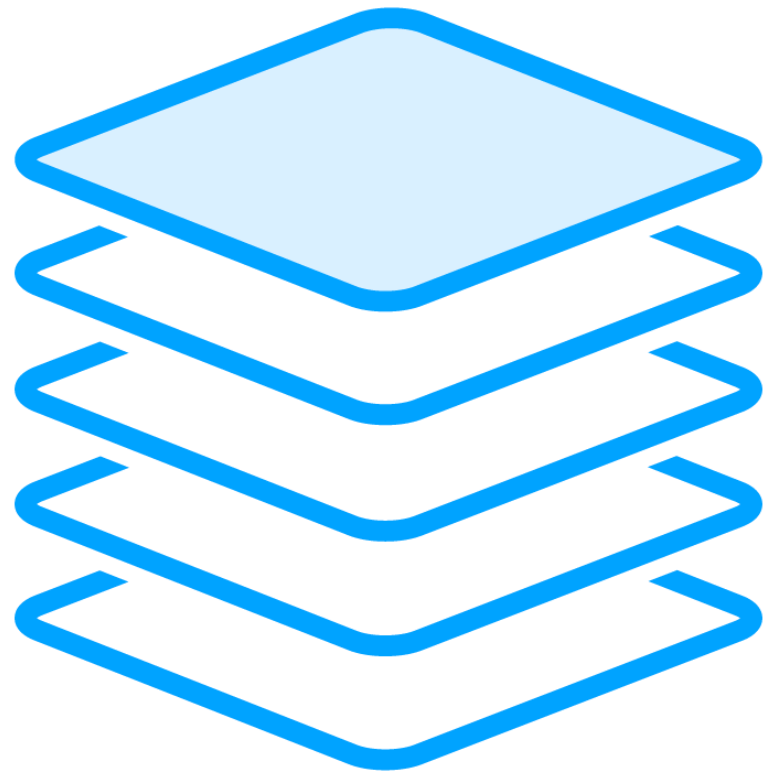
Directories and tfvars files

Branches or release tags

Separate repositories



Terraform OSS Workspaces



Different from Terraform Cloud workspaces

Separate state data per workspace

Default workspace during init

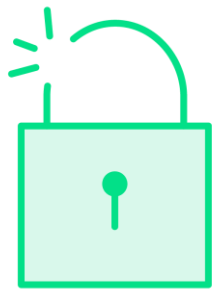
- terraform workspace command

terraform.workspace expression

Short-lived environments for testing



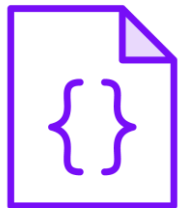
Open-source Workspace Challenges



Shared state data backend without native access controls



Testing and promoting changes



Variable value management



```
.
├── environments
│   ├── development
│   │   ├── main.tf
│   │   ├── development.tfvars
│   │   └── backend.tf
│   ├── production
│   │   └── main.tf
│   └── staging
│       └── main.tf
└── modules
    ├── network
    └── security
```

◀ **Environment directory**

◀ **Development environment**

◀ **Development variable values**

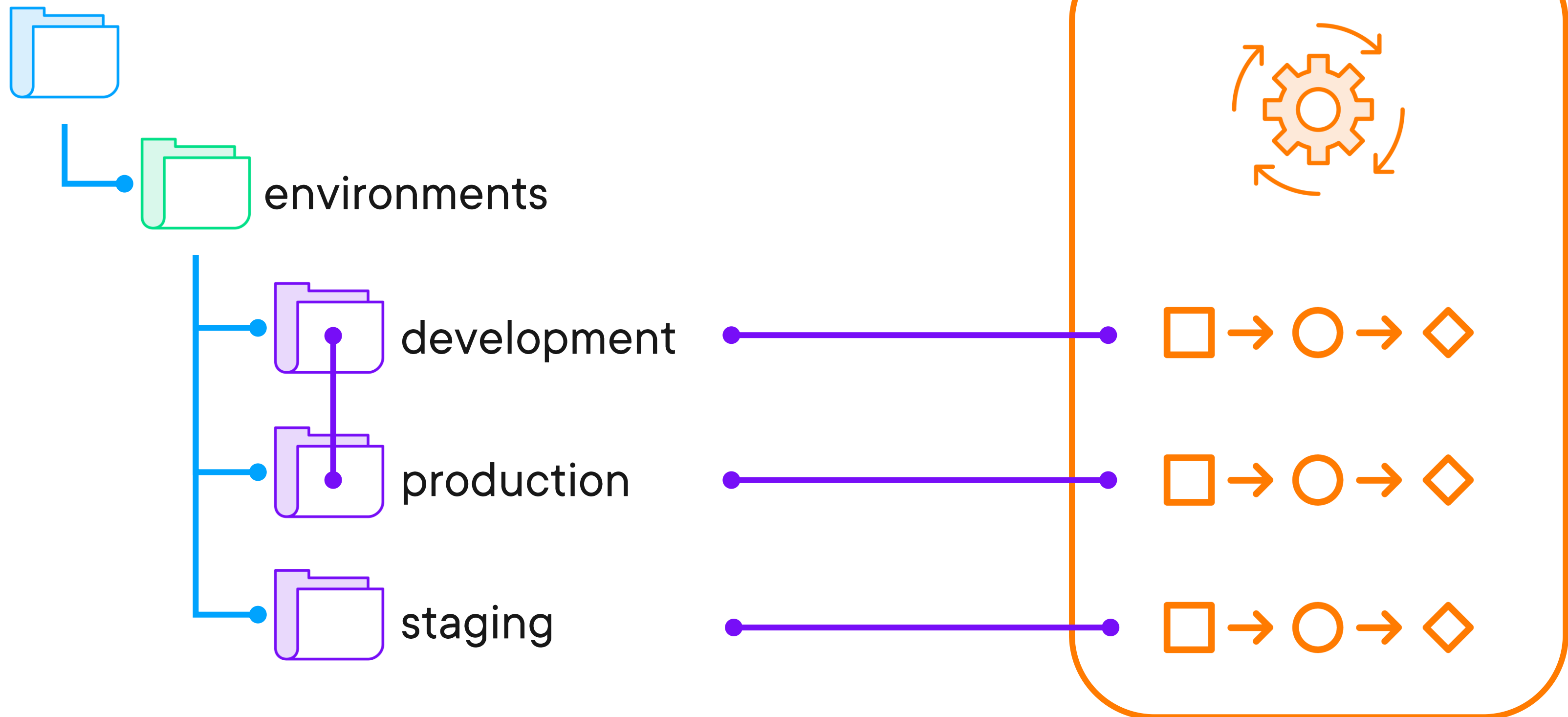
◀ **Development specific backend**

◀ **Child modules for all environments**

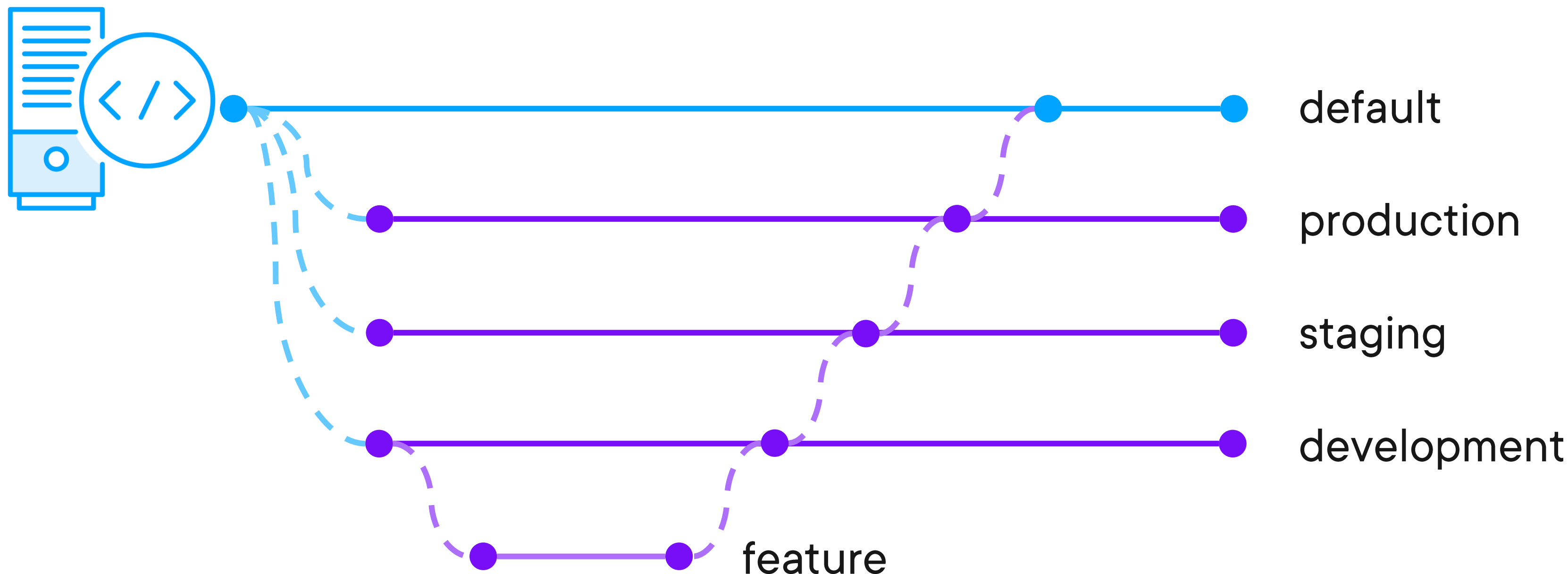
◀ **Move to separate repository and publish**



Separate Directories Workflow



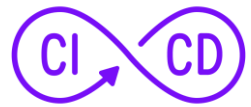
Separate Branches



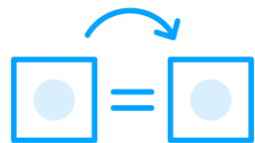
Separate Branches

{JSON}

Same state data backend with different settings



Store variable values in CI/CD pipeline



Environments nearly identical



Requires discipline and operational maturity



Separate Repositories

**True separation of
environments**

**Increased
management
overhead**

**Different state
backends**

**Store variable values
with code**

**Credentials stored
with CI/CD pipeline**



Globomantics Environment



GLOBOMANTICS

Branch-based approach for environments

Terraform Cloud workspaces

- Access controls
- Separate state data
- Variable values and credentials
- Linked to VCS branch

Secure default and production branches





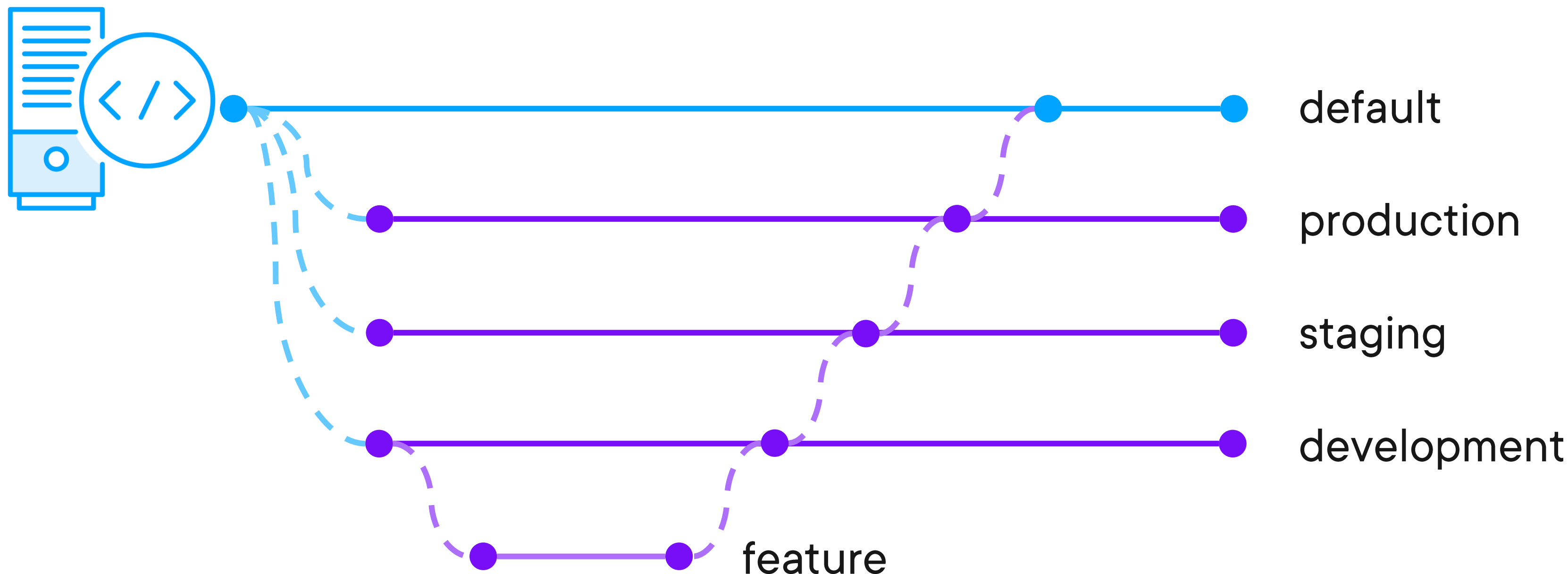
Move workspace to development branch

Process

- Create development branch
- Update workspace settings
- Standard run to validate



Change Promotion





Create two more branches

- Staging
- Production

Create workspaces for each branch

Create a change and promote it

- Add workspace tag



Module Summary



**Git branches and Terraform Cloud
workspaces**

Approaches for multiple environments

Code promotion with feature branches



Up Next:

Connecting Multiple Configurations

